AML TRANSACTION AND FRAUD ANALYSIS

Interactive Dashboard Using Microsoft Power BI

PROJECT REPORT

Submitted By:

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Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps me organize data and present important information through interactive dashboards, charts and KPI cards.

For this project, I created an AML Transaction and Fraud Analysis Dashboard using Microsoft Power BI. I used transaction, fraud-status, account and alert information to examine transaction activity and identify patterns related to fraudulent transactions.

The dashboard presents transaction volume across timestamps, transaction value by fraud status, the distribution of transactions by fraud status, fraudulent transaction volume across timestamps and alert counts by alert type. I also created an interactive page with filters and a separate AML Tooltip page.

2. Objectives of the Project

The main objectives of my project are:

- To monitor the overall number and value of transactions.
- To analyse transaction volume across timestamps.
- To compare total transaction value for fraudulent and non-fraudulent transactions.
- To examine the distribution of transactions by fraud status.
- To review fraudulent transaction volume across timestamps.
- To compare alert counts across alert types.
- To use fraud-status and alert-type filters for interactive analysis.
- To display key transaction and fraud measures through a tooltip page.

3. Dataset Description

I used a dataset containing transaction, account, fraud-status and alert-related information. The visible dashboard fields include timestamp, fraud status, alert type and alert ID. The summary cards display total transactions, total transaction amount, total accounts and fraudulent transactions.

The main fields and measures used in my dashboard are:

Field / Measure	Description
Timestamp	Used to display transaction activity and fraudulent transaction volume over the sequence of timestamps.
Fraud Status	Separates records marked False and True for the fraud-status comparisons.
Transaction Volume	Shows the number or volume of transactions represented across timestamps.
Total Transaction Value	Displays the monetary value of transactions, including the comparison by fraud status.
Alert Type	Groups alerts into the visible types cycle and fan_in.
Alert ID	Used as the counted field in the alert-by-alert-type chart.
Total Accounts	Displays the account count in the dashboard summary.
Fraudulent Transaction	Displays the count of transactions classified as fraudulent.

4. Data Preparation

I organized the transaction information so that I could compare transaction activity across timestamps and separate the records by fraud status. I also used alert type to group alerts and alert ID as the counted field in the alert comparison.

For the time-based visuals, the timestamp field needs to be arranged in the correct chronological order. Fraud status and alert type are treated as categories, while transaction volume, transaction value and alert counts are used as numerical measures. This arrangement supports the comparisons shown in the dashboard.

5. KPI Cards and Summary Measures

I used KPI cards to summarize the main transaction and account measures. The dashboard displays:

- **Total Transactions:** approximately 1M.
- **Total Transaction Amount:** 153.48bn.
- **Total Accounts:** approximately 10K.
- **Fraudulent Transaction:** approximately 2K.

The AML Tooltip page repeats these three transaction-focused measures: Total Transaction of 1M, Total Transaction Amount of 153.48bn and Fraudulent Transaction of 2K. These cards provide a compact summary of the main values used in the analysis.

6. Dashboard Components

I created several visualizations to analyse transaction activity and fraud-related information from different perspectives.

6.1 Transaction Volume Across Timestamp

I used a line chart to show transaction volume across the timestamp sequence. The line begins at approximately 5K, rises quickly to above 6K and then fluctuates mostly between about 6K and 7K. The chart shows repeated short-term rises and falls rather than a steady upward or downward trend.

Purpose: To monitor transaction-volume changes across timestamps and identify periods with relatively higher or lower activity.

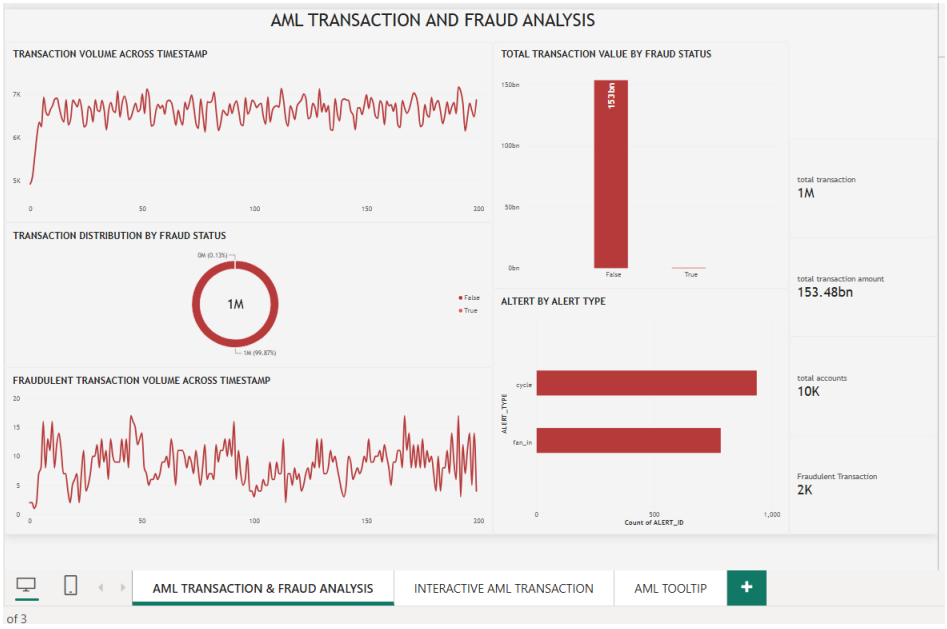


Figure 1: Transaction volume, fraud-status value comparison and fraud distribution.

6.2 Total Transaction Value by Fraud Status

I used a column chart to compare total transaction value for the two fraud-status groups, False and True. The False category accounts for almost all of the displayed transaction value, at approximately 153bn. The True category is very small in comparison and appears close to the baseline.

Purpose: To compare the monetary value associated with transactions classified under each fraud-status value.

6.3 Transaction Distribution by Fraud Status

I used a doughnut chart to show how transactions are distributed between False and True fraud-status categories. The chart indicates approximately 99.87% for False and 0.13% for True. This means the True category represents a very small share of the total transaction distribution shown.

Purpose: To understand the proportion of transactions in each fraud-status group.

6.4 Fraudulent Transaction Volume Across Timestamp

I used a line chart to display fraudulent transaction volume across timestamps. The values fluctuate throughout the displayed sequence, generally remaining below 20. There are several short peaks, including visible rises around the middle and later portions of the sequence, followed by drops. The chart does not show a consistent upward or downward pattern.

Purpose: To examine how the volume of fraudulent transactions changes across timestamps and identify periods with higher values.

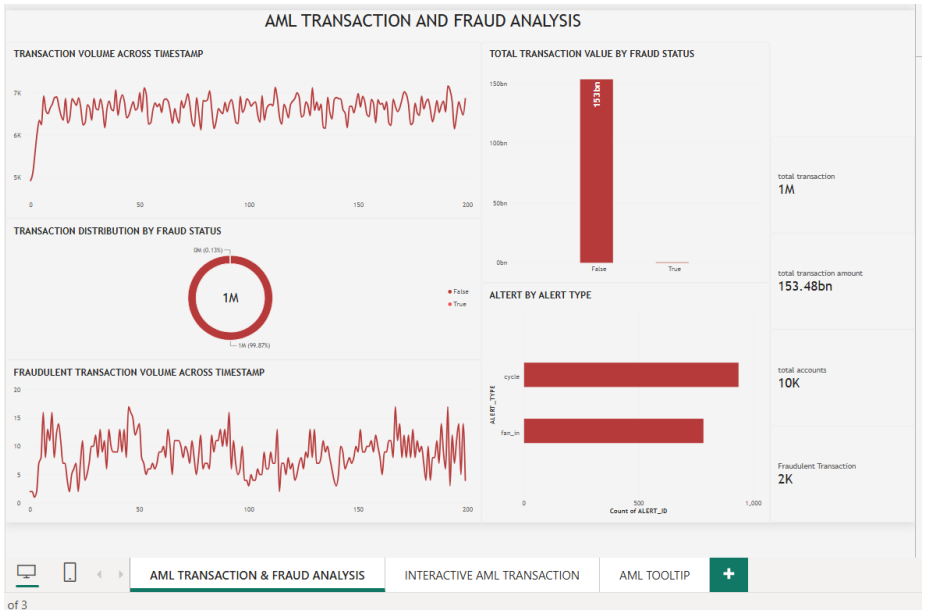


Figure 2: Fraudulent transaction volume and alert counts by alert type.

6.5 Alert by Alert Type

I used a horizontal bar chart to compare the count of Alert_ID across the visible alert types, cycle and fan_in. The cycle bar is longer and reaches approximately 950, while fan_in reaches around 800. Therefore, cycle has the higher alert count among the two types shown.

Purpose: To compare alert frequency by alert type and identify which type appears more often in the displayed data.

7. Interactive Dashboard Filters

I added two slicers to the interactive AML dashboard:

- **Fraud Status:** Allows me to select False or True and focus the report on one fraud-status group.
- **Alert Type:** Allows me to select cycle or fan_in and examine the related alert activity.

These slicers help me narrow the dashboard to selected fraud-status and alert-type values. The visuals can then be reviewed according to the selected filters.

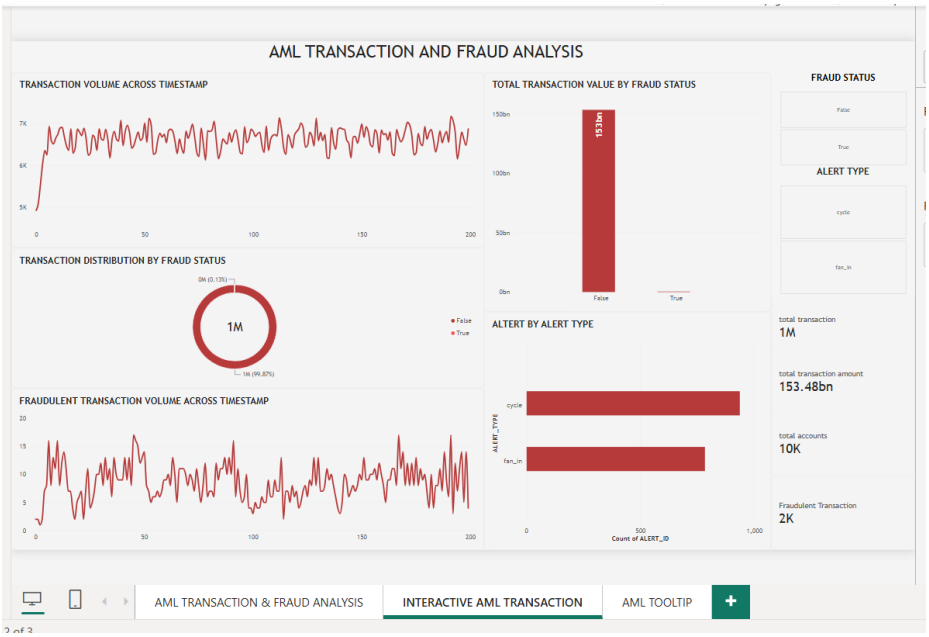


Figure 3: Interactive AML dashboard with Fraud Status and Alert Type slicers.

8. Key Findings

From my dashboard analysis, I identified the following findings:

- The dashboard displays approximately 1M total transactions.
- The total transaction amount is approximately 153.48bn.
- The dashboard displays approximately 10K total accounts and 2K fraudulent transactions.
- Transaction volume fluctuates mostly between approximately 6K and 7K across the displayed timestamp sequence.
- The False fraud-status group accounts for approximately 99.87% of the transaction distribution, while True accounts for approximately 0.13%.
- The total transaction value for False is much higher than the value for True.
- Fraudulent transaction volume varies across timestamps, with visible values generally below 20.
- The cycle alert type has a higher displayed alert count than fan_in.

These findings describe the patterns visible in the dashboard. The charts alone do not establish why a transaction was classified as fraudulent or whether an alert represents a confirmed fraud case.

9. Importance of the Dashboard

- **Transaction Monitoring:** The timestamp line chart helps me review changes in transaction volume.
- **Fraud-Status Comparison:** The column and doughnut charts show differences in transaction value and distribution by fraud status.
- **Fraud Activity Review:** The fraudulent transaction trend helps me examine changes across timestamps.
- **Alert Analysis:** The alert-type chart helps me compare the frequency of cycle and fan_in alerts.
- **Interactive Analysis:** Fraud Status and Alert Type slicers allow me to focus on selected categories.
- **Quick Summary:** KPI cards provide a compact view of transaction count, transaction amount, account count and fraudulent transactions.

10. AML Tooltip

I created a separate AML Tooltip page containing three KPI cards: Total Transaction, Total Transaction Amount and Fraudulent Transaction. The displayed values are approximately 1M, 153.48bn and 2K respectively.

When this page is assigned as a tooltip to a related visual in Power BI, it can display these summary measures when I hover over a data point. This provides additional transaction and fraud information without leaving the main report.

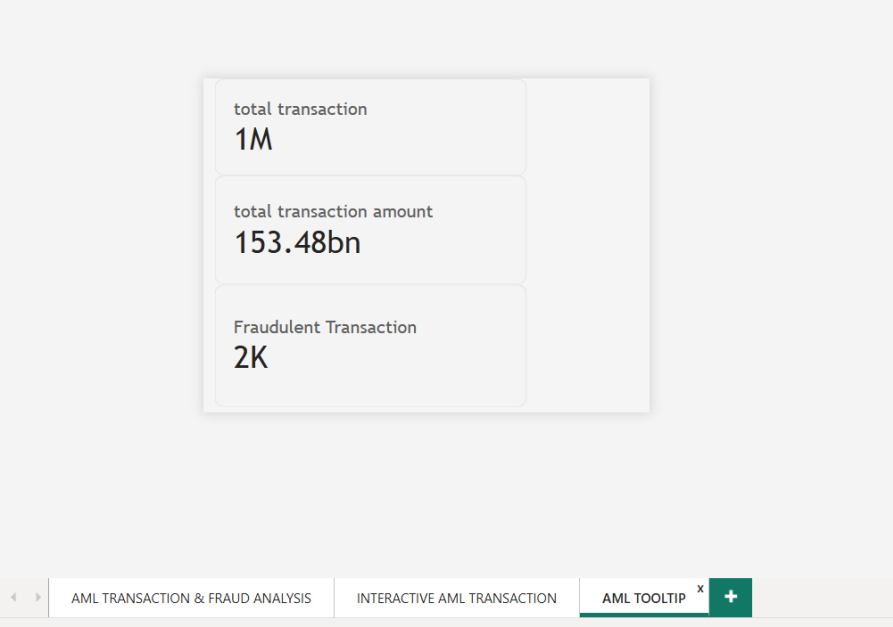


Figure 4: AML Tooltip page with transaction and fraud summary cards.

11. Conclusion

Through this project, I learned how to use Microsoft Power BI to present and analyse anti-money laundering transaction data. I created visuals to monitor transaction volume over timestamps, compare transaction value by fraud status, review the distribution of fraud-status categories and examine fraudulent transaction volume.

I also compared alert counts for the cycle and fan_in types and added slicers to explore the dashboard by fraud status and alert type. The KPI cards and AML Tooltip page provide quick access to the main transaction, account and fraud measures.

Overall, this project improved my understanding of transaction monitoring dashboards, fraud-status analysis, alert comparisons, interactive filtering and tooltip design in Power BI. The dashboard supports review of the patterns in the data, while further investigation would be needed to determine the reasons behind individual fraud classifications or alerts.

12. References

- Microsoft Power BI Desktop.
- AML transaction and fraud analysis dataset used for the project.
- Microsoft Power BI documentation.